Project Check-in #3 reflection

Introduction:

This project explores how structured demographic features influence matchmaking patterns by predicting ideal partner attributes from an individual's profile. Inspired by sociological studies and recent advances in deep learning for tabular data, we apply the TabTransformer architecture to learn contextual relationships among features such as age, education, occupation, and meeting context. The task is framed as a multi-label, multi-class classification problem.

Challenges:

What has been the hardest part of the project you've encountered so far?

1. Feature Extraction Consistency:
    - One key challenge lies in extracting consistent and reliable features using GPT-4 from unstructured text. Fields such as "undergrad" and "how they first met" yielded noisy or inaccurate results. For instance, the undergrad field was inconsistently labeled and may benefit from keyword-based rules rather than relying solely on GPT-generated outputs. Similarly, "how they met" lacked clear extraction criteria and needed a robust heuristic.

2. Categorization Ambiguity:
    - Some attributes (e.g., graduate school, job level) have sparse or ambiguous values. Binary representation was considered for graduate school status, but missing values and vague language ("continued education" or "earned a degree") posed difficulty. Moreover, the categorization of meeting types (e.g., "bar", "church", "friends") required manual grouping that introduces potential bias.

3. Data Imbalance & Mislabeling:
    - Age bins had anomalies (e.g., 12-year-olds), likely due to OCR or GPT misinterpretation. Also, race and religion fields had many missing or unreliable values. These inconsistencies limited the number of clean samples usable for model training.

Insights & Preliminary Results

Model Performance:

- Current accuracy: 43.1% (vs. ~25% random baseline for 4–5 classes).

```
Average Accuracy: 0.4772
second_partner_gender: 0.8357
second_partner_age: 0.2118
second_partner_school_category: 0.5338
second_partner_level_id: 0.4383
second_partner_field: 0.3663
```

- Loss steadily decreased during training (0.27 → 0.16), suggesting the model is learning but may benefit from longer training or architectural tweaks.
- Key limitation: Performance varies significantly by feature (e.g., age predictions may be more reliable than occupation).

Validation:

- Example prediction for [Female, 25-30, Ivy, S2, Business]:

```
=== Predicted Partner Profile ===
Gender:         Male
Age Group:      31
School Category: Ivy League / Elite Private
Education Level: S3
Field of Study:  Business and Financial Occupations
```

Observations: Some logical patterns emerge (e.g., age proximity), but field/education predictions need improvement.

Revised Plan & Next Steps

We are on track with the implementation, but still fine-tuning the preprocessing pipeline and classification heads. The following are key next steps:

- **Refine Preprocessing**:
  - Replace GPT-only extraction for `undergrad` with hybrid keyword matching.
  - Drop or encode unreliable features (e.g., birthplace, race) more selectively.
  - Ensure age is always binned appropriately and filter unrealistic values (e.g., age < 18).
- **Improve Label Binning**:
  - Re-bin age into 5-year groups after verifying clean distribution.
  - Redefine categories for "how they first met" to minimize ambiguity.
- **Model Adjustments**:
  - Explore weighting different output heads based on label quality and cardinality.

    - Compare TabTransformer with simpler baselines (e.g., MLP or XGBoost) as a sanity check.
- **Final Experiments**:
    - Perform ablation studies (e.g., input without occupation).
    - Run more epochs with early stopping and tuned dropout rates.
        - Early Stopping: Run 100+ epochs with patience=10 and LR scheduling.
    - Generate confusion matrices per field to analyze systematic errors.
- **Potential Pivots:**
    - If TabTransformer underperforms baselines, simplify to a multi-output MLP with embeddings.
    - Focus on top 3–4 most reliable features if others remain too noisy.